

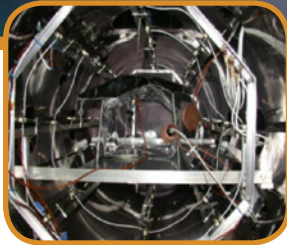


Marshall Space Flight Center Environmental Test Facility

Engineering Solutions for Space Science and Exploration



V20 Vacuum Chamber.



Thermal Vacuum Bakeout.



Shroud Installation in V20.



External Tank Foam Testing.



Thermal Vacuum Chamber.

Environmental Test Facilities (ETF)

Environmental Test Facilities (ETF) are used to simulate the extreme environments that hardware is expected to withstand during operations. The ETF provides these environments using 18 chambers capable of high vacuums, extreme temperatures and humidity, and high-altitude conditions. Additionally, the ETF can perform vacuum bakeout and launch and re-entry testing. Feed throughs in chamber walls provide electrical, mechanical, and fluid connections to the test article.

Thermal Vacuum Testing

The ETF is equipped to perform thermal vacuum tests in any of 8 different chambers. These chambers range in size from 20 x 28 ft to 2 x 2.5 ft. The largest of the ETF chambers, 20 ft diameter.

Thermal-Humidity-Altitude Testing

Environmental simulations of extreme temperatures, humidity, and high-altitude pressures can be achieved in one of the thermal-humidity- altitude test chambers. Thermal conditions alone or thermal conditions and controlled humidity can be simulated in these test chambers. Atmospheric flight conditions with extreme temperature, low pressures, and controlled humidity are simulated in the thermal altitude chambers. The thermal humidity and thermal altitude chambers are cost efficient tools for research of the thermal stability of materials. Instrumentation includes thermocouples, humidity sensors, and pressure gauges.

Life-Cycle Testing

The performance of hardware in space over decades can be simulated in a long-term thermal vacuum test. Two of the thermal vacuum chambers are used for life-cycle testing of space hardware. One chamber is equipped with vacuum rated motors that can position a test article in a liquid nitrogen shroud for rapid cooling. The chamber then has the capability to reposition the article near an array of infrared lamps for rapid heating.



Sunsport Vacuum Chamber.

Optical Cleanliness Thermal Vacuum Bakeout

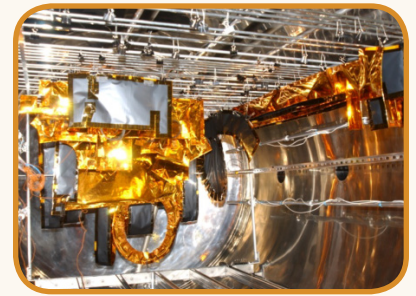
The ETF is recognized as a leader in optical cleanliness vacuum bakeout, meeting MSFC Specification 1238, one of the most stringent certifications of optical cleanliness. The ETF has been used to certify the cleanliness of components used on the Chandra X-Ray Telescope, Hubble Space Telescope, Solar-B Telescope and James Webb Space Telescope. The entry doors of two out of three chambers are inside an ISO Class 7 clean room and the entry of the third chamber is in an ISO Class 8 clean room.

Launch Simulation Testing

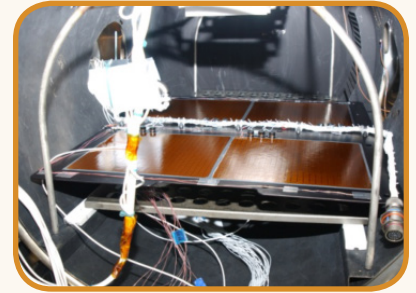
During launch, any flight hardware exposed to the external environment will undergo a rapid pressure change from one atmosphere to a high vacuum within minutes. Likewise, there is a rapid pressure increase upon reentry into the Earth's atmosphere. This facility has the capability to accurately simulate the rapid dynamic depressurization and repressurization experienced during launch and reentry. In addition, two chambers are equipped with cryogenic systems and banks of high-energy infrared lamps that can provide temperature extremes of -240 °C to +343 °C.

Vacuum Bakeout

Thermal vacuum bakeouts are performed in six chambers with test article sizes up to 15-ft diameter. Instrumentation includes thermocouples and convection and ionization pressure gauges.



MLI Blanket Bakeout
in Vacuum Chamber V7.



Heater Mat Assembly Test
in Vacuum Chamber V7.

Chamber Specifications

Chamber	Primary Use	Vacuum Pressure	Temperatures	Thermal Conditioning	Dimensions
BC-1	Vacuum Bakeout	1x10 ⁻⁶ Torr	Ambient to 200°C	IR Lamps	4 ft X 5 ft
V1	Thermal Vacuum / Optical Cleanliness	5x10 ⁻⁷ Torr	Ambient to 180°C	IR Lamps	4 ft X 7 ft
V3	Thermal Vacuum / Life Cycle Testing	5x10 ⁻⁷ Torr	-100 to 100°C	IR Lamps, LN ₂	4 ft X 7.5 ft
V4	Vacuum Bakeout	1x10 ⁻⁶ Torr	Ambient to 175°C	IR Lamps	2 ft X 2.5 ft
V5	Thermal Vacuum	1x10 ⁻⁶ Torr	-170 to 150°C	IR Lamps, LN ₂	3 ft X 4 ft
V6	Thermal Vacuum	1x10 ⁻⁶ Torr	-170 to 150°C	IR Lamps, LN ₂	3 ft X 4 ft
V7	Thermal Vacuum	8x10 ⁻⁷ Torr	-170 to 150°C	IR Lamps, LN ₂	8 ft X 10 ft
V8	Vacuum Bakeout	1x10 ⁻⁶ Torr	Ambient to 175°C	IR Lamps	2 ft X 2.5 ft
V9	Launch Simulation Vacuum	Launch to 100,000 ft or Vacuum to 1X10 ⁻⁶	Ambient to 170°C	IR Lamps, LN ₂	4 ft X 5 ft
V15	Thermal Vacuum	8x10 ⁻⁷ Torr	-170 to 180°C	IR Lamps, LN ₂	12 ft X 20 ft
V16	Altitude	Ambient to 100,000 ft	-170 to 160°C	IR Lamps, LN ₂ or External Chiller	4 ft X 16 ft
Sunspot	Thermal Vacuum	1x10 ⁻⁶ Torr	-170 to 200°C	IR Lamps, LN ₂	10 ft X 12 ft
V20	Thermal Vacuum / Lunar Surface Simulation	5x10 ⁻⁷ Torr	-170 to 200°C	IR Lamps, LN ₂	20 ft X 28 ft
TA1	Thermal Altitude	Ambient to 100,000 ft	-70 to 170°C	Electrical Resistive and Refrigeration	4 ft X 4 ft X 4ft H
TH4	Thermal Humidity	Ambient	-70 to 170°C	Electrical Resistive and Refrigeration	4 ft X 5 ft X 4ft H
TH5	Thermal Humidity	Ambient	-70 to 170°C	Electrical Resistive and Refrigeration	4 ft X 4 ft X 4ft H
TH6	Thermal Humidity	Ambient	-70 to 170°C	Electrical Resistive and Refrigeration	4 ft X 4 ft X 4ft H
TH7	Thermal Humidity	Ambient	-70 to 170°C	Electrical Resistive and Refrigeration	4 ft X 4 ft X 4ft H

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NP-2021-03-08-MSFC